# D-Flow / QGF
# Overnight investigation

## Third campaign | November 6-7, 2026
## LAYOUT DRAFT - RESULTS STILL IN PROGRESS

Preserved40 evidence cutoff: 2026-11-07T08:11:35.235744+00:00
Main target100 evidence cutoff: 2026-11-07T11:18:22.743822+00:00

The current comparison uses four D-Flow families: regular or stochastic, each with raw or nearby-action averaged Q. Each family has a separate λ0 and λ1 condition, giving eight D-Flow conditions. All use identity-Jacobian guidance and complete eight-slice flow endpoints.

The controls are BC, original QGF at fixed κ=.02/.04, independently tuned original QGF, one original IPG and one original FK corrector. The main target is 100 heldout episodes per unique frozen configuration and trained model, with training seeds 11, 23 and 37. All 84 primary cells completed 100 episodes each in the supplied closed snapshot. Older separately declared 40-episode panels remain 40.

| Closed heldout evidence in this snapshot | Episode rows |
|---|---|
| Task1/5 main fresh100, new episodes | 8400 |
| Earlier separate 40-episode panels, new episodes | 600 |
| Task2 locked controls fresh100, new episodes | 1200 |
| Total new heldout episodes across separate protocols | 10200 |
| Inherited complete A20 reference episodes | 200 |
| Training updates in this campaign | 0 |

These counts deduplicate main A/B references across the 40 and 100 snapshots. The separate Task2 control study adds 1200 heldout episodes; its locked historical QGF strengths do not complete a full Task2 D-Flow/current-domain sweep. Canonical-TRAIN Monte Carlo diagnostics, validation and offline engineering checks are separate from this 10200 heldout total. Same-start comparisons are paired observations; 100 starts do not create 100 independently trained models.

This report supplements the preserved earlier campaigns. Historical synthetic points, custom navigation, coverage-stress navigation and cube manipulation must not be pooled with the standard native episodes here.

## What does the agent actually do?

Tasks 1 and 5 use native MuJoCo PointMaze navigation with real state transitions. A point agent selects two-dimensional control actions to reach the task goal while navigating the maze. This is navigation, separate from the earlier cube-manipulation study.

Task 1 navigates toward nominal goal (20,20). Task 5 starts near nominal (20,4) and navigates toward (0,0). Resets may vary within the task-defined start region. The standard evaluation cap is 1000 control decisions, separate from the earlier modified 128-step coverage test.

BC and IQL have different jobs. BC learns the offline action distribution. IQL learns Q(s,a) and V(s) from offline transitions including next states. This campaign freezes the BC actor from update 1000 and the IQL critic from update 50000; samplers propose actions but do not retrain Q or V.

The guidance score is the minimum of the two frozen target-Q heads. It is an estimate of future task return, not a certificate that a proposed action is safe or well supported by the offline data. “Unseen Q” here means evaluating this learned critic on actions outside or poorly covered by that data.

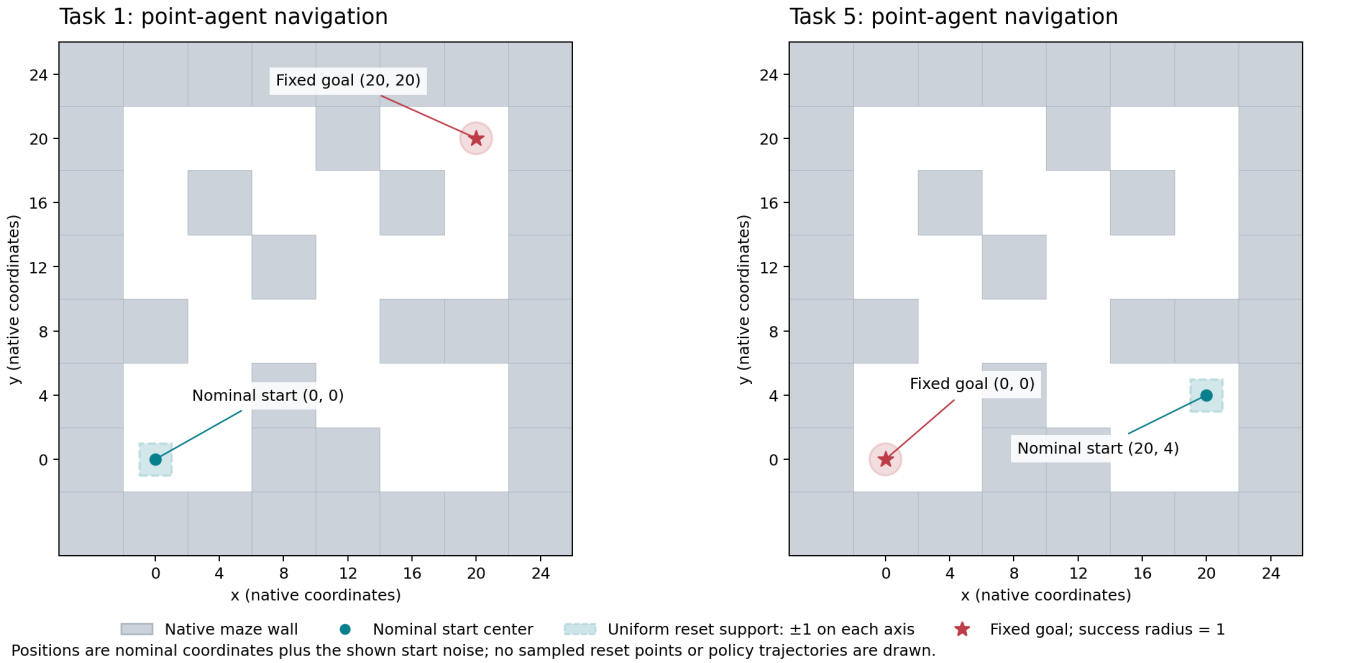
The native sparse reward is success minus one: -1 outside the radius-1 goal region, 0 on success. Success is checked before the action. The logged current position determines the relabeled reward and mask; IQL uses y=r+.99(1-success)V(actual next state), with V expectile .8. Larger Q predicts better discounted task return. The BC flow itself is ordinary behavior matching, with position input and no goal vector.

Seeds 11/23/37 identify independently trained actor/critic models. They are not task numbers or evaluation reset IDs. All three models remain in the report, including weak or failed ones.

# Official maze geometry and task goals

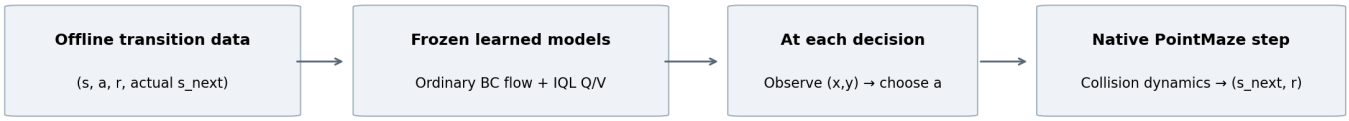
## What the native experiments ask the agent to do

Navigate the point-agent around the actual walls to its fixed goal. Same official medium maze; different start and task reward.




Positions are nominal coordinates plus the shown start noise; no sampled reset points or policy trajectories are drawn.

Native task budget: up to 1,000 control decisions. Success is checked before the action; reward is −1 before success and 0 at success.




BC proposes actions; the fixed IQL-trained critic guides them. Q estimates discounted task reward and can be wrong; it is not an oracle.

These are sequential point-agent experiments with actual next states. The actor has 1,000 BC updates; the task-specific Q/V has 50,000 IQL updates.

Separate historical studies used a 128-decision TRAIN-anchor coverage stress or cube manipulation. Those resets, budgets and robot objectives differ.

Source-only drawing: official map/task literals and coordinate transform verified; no environment creation, reset, model call, physics or random draw.

Both tasks use this same official medium maze. Task1 starts around (0,0) and aims for (20,20); Task5 starts around (20,4) and aims for (0,0). Shaded start boxes show the native ±1 positional reset support, not sampled episodes. Red stars and radius-1 disks mark fixed goals. No policy route is drawn.

The figure comes from source-verified maze/task definitions and the exact coordinate transform. The model input is position (x,y); next states come from native collision dynamics. The flow time slices, D-Flow source iterations and physical control decisions are different time axes.

# What κ, η, λ and averaging mean

Let F8(s,z) be the complete eight-slice frozen BC decoder including final action clipping. Raw S(s,a)=min_h Q_h(s,a). For averaged guidance, Qbar_K(s,a)=(1/K) Σ_k min_h Q_h(s,clip(a+σ ε_k,-1,1)). The head minimum is taken before averaging; K is the total 4 or 16 samples in an antithetic standard-normal bank.

Perturbations are generated around the current candidate action, not retrieved from dataset actions or added as new transition data. The critic remains frozen offline IQL. For example, with action coordinate .20 and an illustrative ε pair ±1, σ=.03 gives .23 and .17 before clipping. σ is the per-coordinate standard deviation before clipping, not a hard radius; other Gaussian draws can lie farther away. The bank is fixed through the outer I updates of one action call and fresh at the next physical decision. This finite-K, bank-conditional score is not the exact Gaussian expectation.

| Operation | Meaning in this campaign |
|---|---|
| κ (kappa) | Multiplier on Q-gradient guidance. Independently selected by task, trained model, method and fixed λ. |
| η (eta) | Step size for each D-Flow source update. |
| λ (lambda) | Coefficient of source-prior pull -λz. λ0 removes this pull; λ1 includes it. |
| I | Number of outer D-Flow source updates. Each update uses a full-flow endpoint. |
| σ / K | Nearby-action perturbation scale and total averaging samples. The antithetic bank stays fixed throughout one action generation. |

Identity guidance: g = grad_a S(s,a), evaluated at a=F8(s,z), without multiplying by the endpoint Jacobian. Regular update: z ← z + η(κg - λz). Stochastic update adds sqrt(2ηT) ξ, with T=1. The executed action is F8(s,z_I).

Regular omits the added outer-source ξ noise. Its initial Gaussian z0, and the Qavg perturbation bank when used, still randomize actions; regular does not mean a deterministic policy.

Setting κ=0 removes Q guidance. It does not necessarily reproduce BC: λ1 still pulls z toward zero, and stochastic source noise still moves z. The unchanged BC flow remains a separate canonical control.

Source noise moves z. Q averaging changes the action score. Neither operation establishes correction of broad critic extrapolation error. Because the identity field is a surrogate, these runs do not imply exact Langevin sampling of Q(F8(z)). MH is outside this campaign’s scope.

Full-flow endpoints integrate all eight BC time slices. The excluded one-step D-Flow endpoint extrapolation is different from the numerical integrator used inside this complete flow. Original QGF still refreshes its own velocity correction and endpoint estimate at every flow time slice. QGF has no D-Flow η, λ or outer I setting.

Original QGF at each t=i/8: v_i=v(s,a_i,t); e_i=clip(a_i+(1-t)v_i); g_i=grad_a Q(s,a) evaluated at e_i; a_(i+1)=a_i+(v_i+κg_i)/8. It recomputes v_i, e_i and g_i at all eight slices, then applies its final action clipping. Its κ multiplies a velocity correction; D-Flow κ acts inside the repeated source update, so equal numerical κ values need not mean equal action changes.

Independent selection domain: κ=0,.02,.04,.06,.08,.1,.2,.3,.4,.6,.8,1,2,3,5,8,12,20,40,80; η=.005,.01,.03,.1; I=1,2,4,8,16,32,64,128,256; σ=.01,.03,.1,.2; K=4,16. Averaged-Q search uses 720 base settings plus at most 70 refinements per condition, followed by 8-episode physical validation. This hierarchy is not an exhaustive σ/K Cartesian sweep.

# Earlier separate 40-episode panels

Historical panel: these separately declared 612xxx cohorts are not the current paired main target100 comparison. The current all-three-model results appear in the main comparison tables below.

# Task 1: original A20 + fresh B20

| Method | Seed 11 | Seed 23 | Seed 37 |
|---|---|---|---|
| BC flow | 0/40 | Pending | Pending |
| QGF fixed .02 | 0/40 | Pending | Pending |
| QGF fixed .04 | 0/40 | Pending | Pending |
| QGF tuned | 29/40 | Pending | Pending |
| IPG | 22/40 | Pending | Pending |
| FK corrector | 19/40 | Pending | Pending |
| DF Regular, raw Q, λ0 | 28/40 | Pending | Pending |
| DF Regular, raw Q, λ1 | 32/40 | Pending | Pending |
| DF Stochastic, raw Q, λ0 | 24/40 | Pending | Pending |
| DF Stochastic, raw Q, λ1 | 24/40 | Pending | Pending |

# Task 5: new A20 + new B20

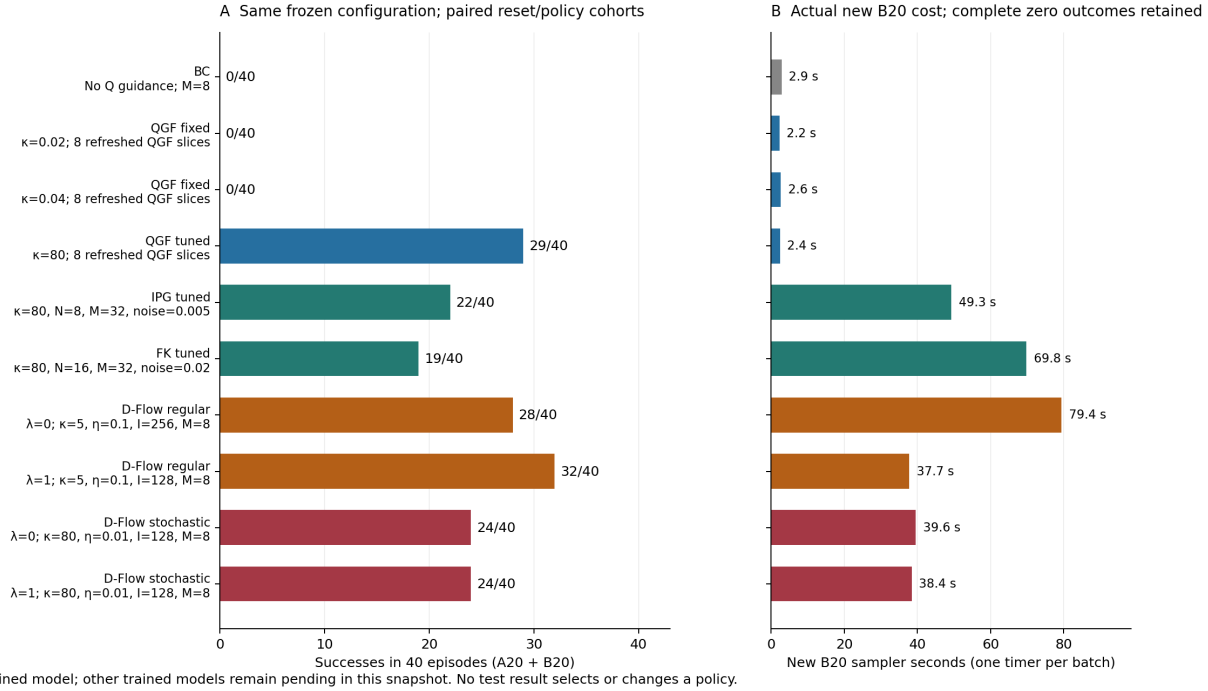
| Method | Seed 11 | Seed 23 | Seed 37 |
|---|---|---|---|
| BC flow | 0/40 | Pending | Pending |
| QGF fixed .02 | 0/40 | Pending | Pending |
| QGF fixed .04 | 0/40 | Pending | Pending |
| QGF tuned | 37/40 | Pending | Pending |
| IPG | 40/40 | Pending | Pending |
| FK corrector | 40/40 | Pending | Pending |
| DF Regular, raw Q, λ0 | 37/40 | Pending | Pending |
| DF Regular, raw Q, λ1 | 40/40 | Pending | Pending |
| DF Stochastic, raw Q, λ0 | 35/40 | Pending | Pending |
| DF Stochastic, raw Q, λ1 | 40/40 | Pending | Pending |

Task1 may reference eligible original A20 and add fresh B20 at unchanged frozen settings. Task5 has its own separately declared new panel. All relevant three-model selections freeze before that branch’s tests. Reported totals require both disjoint cohorts and an identical selected configuration.

# Historical model 11: 40-episode success and cost

Task 1, trained seed 11: frozen raw D-Flow versus QGF, IPG, FK and BC

Full-flow endpoint; identity Jacobian only for D-Flow. Each guided condition keeps its own validation-selected settings.




One independently trained model; other trained models remain pending in this snapshot. No test result selects or changes a policy.

Cost includes padded B20 execution and unequal early stopping, particle/flow counts and compilation effects; it is not a Q/NFE query count.

A20 is inherited evidence; B20 is new. Q-averaged identity methods use a separate fresh panel and are not represented by zero bars here.

This preserved figure shows Task1 training seed11 only: ten frozen policies on its original A20 plus fresh B20. The right panel times only newly executed B20 batches. It does not compare the new averaged-Q panel, which has separate starts and may still be pending.

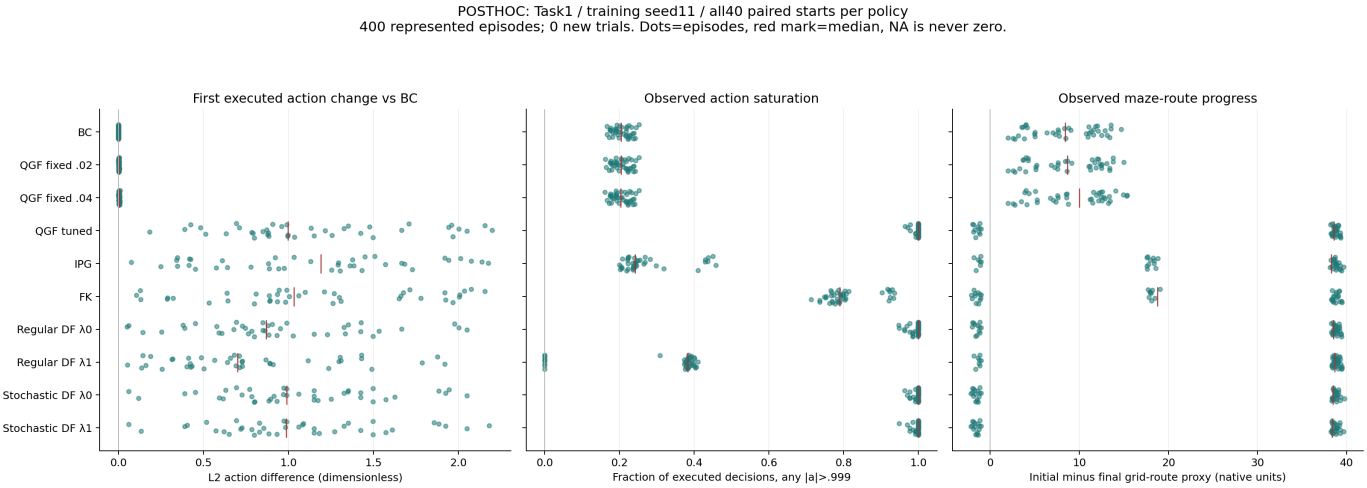
Cost is measured once per batch. It includes unequal inference budgets and early stopping, and can include compilation effects. The three extra successes for regular λ1 relative to tuned QGF in this model do not establish a universal winner or isolate the effect of λ, since the selected iteration counts also differ.

# Posthoc saved-trajectory description - Task 1 / model 11

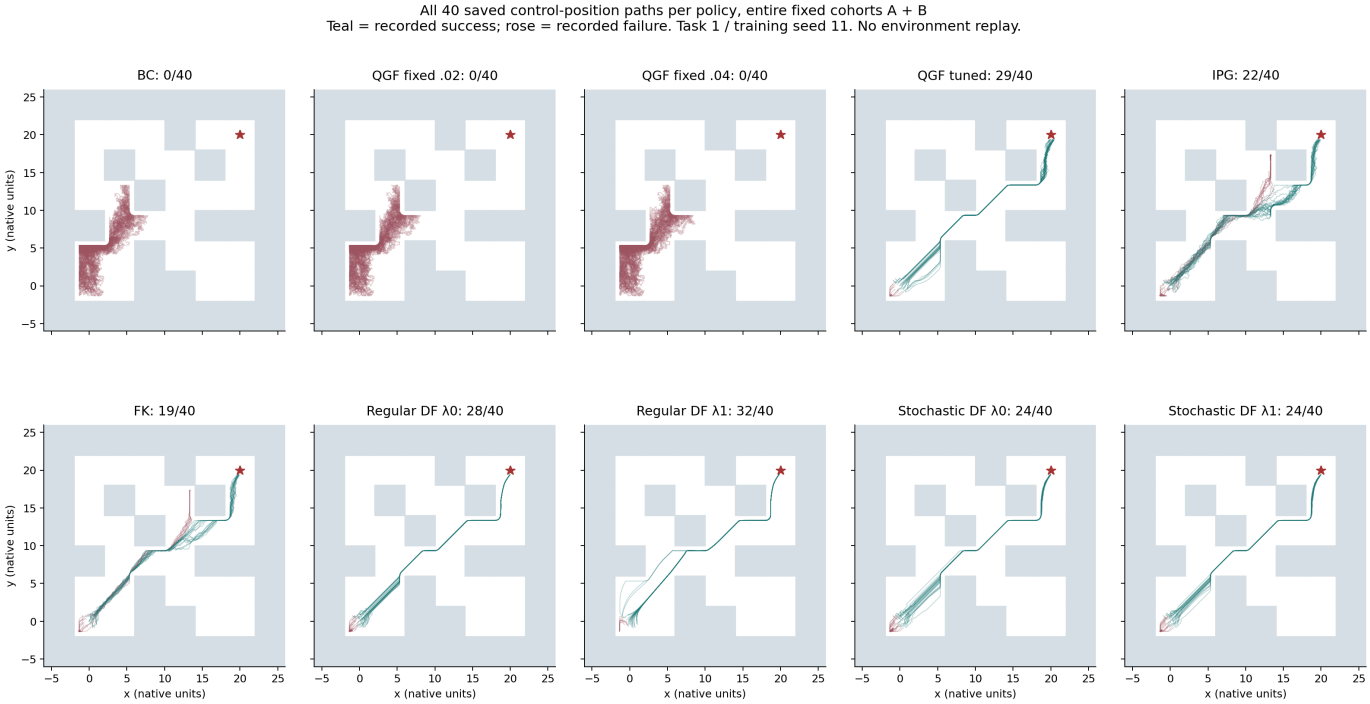
This analysis was specified after outcomes existed. It describes all 400 already represented policy/episode rows from one trained model: 200 inherited A rows and 200 previously executed B rows. It adds 0 trials, model queries or environment steps, and did not select or retune any configuration.

| Method | Success / 40 | Verified first-action pairs | Missing chosen-action Q rows |
|---|---|---|---|
| BC flow | 0 | 40/40 | 0 |
| QGF fixed .02 | 0 | 40/40 | 0 |
| QGF fixed .04 | 0 | 40/40 | 0 |
| QGF tuned | 29 | 40/40 | 0 |
| IPG | 22 | 40/40 | 40 |
| FK corrector | 19 | 40/40 | 40 |
| DF Regular, raw Q, λ0 | 28 | 40/40 | 0 |
| DF Regular, raw Q, λ1 | 32 | 40/40 | 0 |
| DF Stochastic, raw Q, λ0 | 24 | 40/40 | 0 |
| DF Stochastic, raw Q, λ1 | 24 | 40/40 | 0 |

Pairing requires matching initial observation and full native-state hashes, reset/policy/step context and saved source0. Particle0 equality does not match the whole particle ensemble or its selected particle. Own settings and 32 versus 8 flow slices remain confounds. Chosen-action Q is shown only when the exact field was saved; missing is NA. Learned Q is not true value or calibration.



# All saved control-position paths - posthoc Task 1 / model 11




All 40 saved paths per policy are shown, including failures. These connect executed control-decision positions rather than MuJoCo internal substeps. The route measure is a cell/BFS proxy, not an exact continuous geodesic; near-zero displacement is not a calibrated collision test. These descriptive outcomes do not isolate a causal noise, prior or averaging effect.

# Main paired comparison: Task 1, target 100

| Method | Seed 11 observed | Seed 23 observed | Seed 37 observed |
|---|---|---|---|
| BC flow | 0/100 | 0/100 | 0/100 |
| QGF fixed .02 | 0/100 | 0/100 | 0/100 |
| QGF fixed .04 | 0/100 | 0/100 | 0/100 |
| QGF tuned | 79/100 | 100/100 | 100/100 |
| IPG | 60/100 | 100/100 | 92/100 |
| FK corrector | 40/100 | 100/100 | 100/100 |
| DF Regular, raw Q, λ0 | 78/100 | 100/100 | 100/100 |
| DF Regular, raw Q, λ1 | 75/100 | 100/100 | 100/100 |
| DF Stochastic, raw Q, λ0 | 72/100 | 100/100 | 100/100 |
| DF Stochastic, raw Q, λ1 | 72/100 | 100/100 | 100/100 |
| DF Regular, average Q, λ0 | 75/100 | 100/100 | 100/100 |
| DF Regular, average Q, λ1 | 78/100 | 100/100 | 100/100 |
| DF Stochastic, average Q, λ0 | 74/100 | 100/100 | 100/100 |
| DF Stochastic, average Q, λ1 | 70/100 | 100/100 | 100/100 |

All cells have target 100. Each shown fraction uses only completed, explicitly recorded B20 request evidence; a missing or partial cell is not 0/100. The 84 primary cells cover 14 methods × 3 trained models × 2 tasks. Required positive-only requests are reported separately.

The original fresh A20+B20 remain unchanged references; C20,D20,E20 extend the same selected configurations without retuning. All 33 guided choices across three models freeze before the task's first main heldout batch. All three models finish every preceding round before a later round release. Paired methods use the same task reset IDs/policy root; disjoint reset IDs give distinct episode-key contexts, without a pooled-independence guarantee.

The historical 612xxx original/extension 40-episode panel above is a different comparison and is never added to these 100. Display aliases consume no additional physical episode when they reference an existing configID. Actual separate numeric-policy reruns still consume cost.

# Main paired comparison: Task 5, target 100

| Method | Seed 11 observed | Seed 23 observed | Seed 37 observed |
|---|---|---|---|
| BC flow | 0/100 | 0/100 | 0/100 |
| QGF fixed .02 | 0/100 | 0/100 | 0/100 |
| QGF fixed .04 | 0/100 | 0/100 | 0/100 |
| QGF tuned | 96/100 | 24/100 | 100/100 |
| IPG | 98/100 | 65/100 | 100/100 |
| FK corrector | 100/100 | 54/100 | 100/100 |
| DF Regular, raw Q, λ0 | 98/100 | 26/100 | 100/100 |
| DF Regular, raw Q, λ1 | 100/100 | 62/100 | 100/100 |
| DF Stochastic, raw Q, λ0 | 92/100 | 47/100 | 100/100 |
| DF Stochastic, raw Q, λ1 | 97/100 | 33/100 | 100/100 |
| DF Regular, average Q, λ0 | 97/100 | 26/100 | 100/100 |
| DF Regular, average Q, λ1 | 100/100 | 62/100 | 100/100 |
| DF Stochastic, average Q, λ0 | 94/100 | 57/100 | 100/100 |
| DF Stochastic, average Q, λ1 | 94/100 | 31/100 | 100/100 |

All cells have target 100. Each shown fraction uses only completed, explicitly recorded B20 request evidence; a missing or partial cell is not 0/100. The 84 primary cells cover 14 methods × 3 trained models × 2 tasks. Required positive-only requests are reported separately.

The original fresh A20+B20 remain unchanged references; C20,D20,E20 extend the same selected configurations without retuning. All 33 guided choices across three models freeze before the task's first main heldout batch. All three models finish every preceding round before a later round release. Paired methods use the same task reset IDs/policy root; disjoint reset IDs give distinct episode-key contexts, without a pooled-independence guarantee.

The historical 612xxx original/extension 40-episode panel above is a different comparison and is never added to these 100. Display aliases consume no additional physical episode when they reference an existing configID. Actual separate numeric-policy reruns still consume cost.

An illustrative whole-policy comparison is Task5 model23: raw stochastic λ0 achieved 47/100 at κ=5, η=0.1, I=128; averaged stochastic λ0 achieved 57/100 at κ=1, η=0.1, I=256, σ=0.01, K=16. Guidance strength and iteration count differ, so this does not isolate an averaging effect. Regular raw/averaged success counts were 26/26 for λ0 and 62/62 for λ1, each out of 100 at independently selected settings. Some other trained models are near the success ceiling. These results establish neither broad critic repair nor accurate Q on unseen actions.

# Main target100: completed evidence and incomplete work

| Evidence/accounting unit | Observed |
|---|---|
| Main A/B completed physical episode evidence | 3360 |
| Additional C/D/E completed physical episode evidence | 5040 |
| Main canonical completed B20 CSV batches | 420 |
| Main requested display/cohort rows | 420 |
| Main required-positive-secondary declarations | 0 |
| Observed active/interrupted resume paths | 0 |
| Unrequested yet finalized canonical CSVs | 0 |

Across this report’s preserved 40-episode and main 100-episode inputs, each actual physical CSV is counted once: completed new-batch sampler time 14580.259s, completed new-batch wall time 15116.605s. These cumulative batch timers can include compilation and are repeated across episode rows in raw CSVs; they are never multiplied by 20. Unknown timers and unfinished prefix compute are not guessed.

A canonical CSV may be finalized just before its requested display proof is appended. Such real physical evidence and cost are retained separately; a requested result is shown only after its recorded request exists. Volatile .resume.pkl presence is a point-in-time metadata observation, with unknown partial episode/physics/cost counters. No pickle or trajectory arrays are loaded by this report consumer.

# Selected settings (1-21 of 84)

| Task / seed | Requested condition | κ | η / l | λ | σ / K | N / M | noise / ridge |
|---|---|---|---|---|---|---|---|
| 1 / 11 | BC flow | 0 | - | - | - | 1 / 8 | - |
| 1 / 11 | QGF fixed .02 | 0.02 | - | - | - | 1 / 8 | - |
| 1 / 11 | QGF fixed .04 | 0.04 | - | - | - | 1 / 8 | - |
| 1 / 11 | QGF tuned | 80 | - | - | - | 1 / 8 | - |
| 1 / 11 | IPG | 80 | - | - | - | 8 / 32 | 0.005 / 0.1 |
| 1 / 11 | FK corrector | 80 | - | - | - | 16 / 32 | 0.02 / - |
| 1 / 11 | DF Regular, raw Q, λ0 | 5 | 0.1 / 256 | 0 | - | 1 / 8 | - |
| 1 / 11 | DF Regular, raw Q, λ1 | 5 | 0.1 / 128 | 1 | - | 1 / 8 | - |
| 1 / 11 | DF Stochastic, raw Q, λ0 | 80 | 0.01 / 128 | 0 | - | 1 / 8 | - |
| 1 / 11 | DF Stochastic, raw Q, λ1 | 80 | 0.01 / 128 | 1 | - | 1 / 8 | - |
| 1 / 11 | DF Regular, average Q, λ0 | 20 | 0.1 / 256 | 0 | 0.1 / 16 | 1 / 8 | - |
| 1 / 11 | DF Regular, average Q, λ1 | 80 | 0.03 / 256 | 1 | 0.03 / 16 | 1 / 8 | - |
| 1 / 11 | DF Stochastic, average Q, λ0 | 80 | 0.03 / 128 | 0 | 0.1 / 4 | 1 / 8 | - |
| 1 / 11 | DF Stochastic, average Q, λ1 | 80 | 0.03 / 256 | 1 | 0.2 / 4 | 1 / 8 | - |
| 1 / 23 | BC flow | 0 | - | - | - | 1 / 8 | - |
| 1 / 23 | QGF fixed .02 | 0.02 | - | - | - | 1 / 8 | - |
| 1 / 23 | QGF fixed .04 | 0.04 | - | - | - | 1 / 8 | - |
| 1 / 23 | QGF tuned | 80 | - | - | - | 1 / 8 | - |
| 1 / 23 | IPG | 40 | - | - | - | 16 / 32 | 0.02 / 0.01 |
| 1 / 23 | FK corrector | 80 | - | - | - | 16 / 32 | 0.02 / - |
| 1 / 23 | DF Regular, raw Q, λ0 | 40 | 0.1 / 256 | 0 | - | 1 / 8 | - |

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

# Selected settings (22-42 of 84)

| Task / seed | Requested condition | κ | η / I | λ | σ / K | N / M | noise / ridge |
|---|---|---|---|---|---|---|---|
| 1 / 23 | DF Regular, raw Q, λ1 | 80 | 0.03 / 256 | 1 | - | 1 / 8 | - |
| 1 / 23 | DF Stochastic, raw Q, λ0 | 80 | 0.03 / 256 | 0 | - | 1 / 8 | - |
| 1 / 23 | DF Stochastic, raw Q, λ1 | 80 | 0.03 / 128 | 1 | - | 1 / 8 | - |
| 1 / 23 | DF Regular, average Q, λ0 | 40 | 0.1 / 256 | 0 | 0.2 / 16 | 1 / 8 | - |
| 1 / 23 | DF Regular, average Q, λ1 | 80 | 0.03 / 256 | 1 | 0.2 / 4 | 1 / 8 | - |
| 1 / 23 | DF Stochastic, average Q, λ0 | 80 | 0.1 / 256 | 0 | 0.03 / 16 | 1 / 8 | - |
| 1 / 23 | DF Stochastic, average Q, λ1 | 80 | 0.03 / 128 | 1 | 0.2 / 4 | 1 / 8 | - |
| 1 / 37 | BC flow | 0 | - | - | - | 1 / 8 | - |
| 1 / 37 | QGF fixed .02 | 0.02 | - | - | - | 1 / 8 | - |
| 1 / 37 | QGF fixed .04 | 0.04 | - | - | - | 1 / 8 | - |
| 1 / 37 | QGF tuned | 40 | - | - | - | 1 / 8 | - |
| 1 / 37 | IPG | 40 | - | - | - | 16 / 8 | 0.02 / 0.01 |
| 1 / 37 | FK corrector | 80 | - | - | - | 16 / 32 | 0.02 / - |
| 1 / 37 | DF Regular, raw Q, λ0 | 5 | 0.1 / 256 | 0 | - | 1 / 8 | - |
| 1 / 37 | DF Regular, raw Q, λ1 | 80 | 0.03 / 256 | 1 | - | 1 / 8 | - |
| 1 / 37 | DF Stochastic, raw Q, λ0 | 80 | 0.1 / 256 | 0 | - | 1 / 8 | - |
| 1 / 37 | DF Stochastic, raw Q, λ1 | 80 | 0.03 / 128 | 1 | - | 1 / 8 | - |
| 1 / 37 | DF Regular, average Q, λ0 | 40 | 0.1 / 128 | 0 | 0.2 / 4 | 1 / 8 | - |
| 1 / 37 | DF Regular, average Q, λ1 | 80 | 0.03 / 256 | 1 | 0.2 / 4 | 1 / 8 | - |
| 1 / 37 | DF Stochastic, average Q, λ0 | 80 | 0.03 / 128 | 0 | 0.2 / 4 | 1 / 8 | - |
| 1 / 37 | DF Stochastic, average Q, λ1 | 80 | 0.03 / 128 | 1 | 0.2 / 4 | 1 / 8 | - |

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

# Selected settings (43-63 of 84)

| Task / seed | Requested condition | κ | η / l | λ | σ / K | N / M | noise / ridge |
|---|---|---|---|---|---|---|---|
| 5 / 11 | BC flow | 0 | - | - | - | 1 / 8 | - |
| 5 / 11 | QGF fixed .02 | 0.02 | - | - | - | 1 / 8 | - |
| 5 / 11 | QGF fixed .04 | 0.04 | - | - | - | 1 / 8 | - |
| 5 / 11 | QGF tuned | 20 | - | - | - | 1 / 8 | - |
| 5 / 11 | IPG | 20 | - | - | - | 16 / 32 | 0.005 / 0.01 |
| 5 / 11 | FK corrector | 80 | - | - | - | 8 / 32 | 0.02 / - |
| 5 / 11 | DF Regular, raw Q, λ0 | 1 | 0.1 / 256 | 0 | - | 1 / 8 | - |
| 5 / 11 | DF Regular, raw Q, λ1 | 5 | 0.1 / 128 | 1 | - | 1 / 8 | - |
| 5 / 11 | DF Stochastic, raw Q, λ0 | 80 | 0.1 / 256 | 0 | - | 1 / 8 | - |
| 5 / 11 | DF Stochastic, raw Q, λ1 | 5 | 0.03 / 128 | 1 | - | 1 / 8 | - |
| 5 / 11 | DF Regular, average Q, λ0 | 1 | 0.1 / 256 | 0 | 0.01 / 16 | 1 / 8 | - |
| 5 / 11 | DF Regular, average Q, λ1 | 5 | 0.1 / 256 | 1 | 0.01 / 16 | 1 / 8 | - |
| 5 / 11 | DF Stochastic, average Q, λ0 | 40 | 0.005 / 256 | 0 | 0.01 / 4 | 1 / 8 | - |
| 5 / 11 | DF Stochastic, average Q, λ1 | 80 | 0.005 / 256 | 1 | 0.01 / 16 | 1 / 8 | - |
| 5 / 23 | BC flow | 0 | - | - | - | 1 / 8 | - |
| 5 / 23 | QGF fixed .02 | 0.02 | - | - | - | 1 / 8 | - |
| 5 / 23 | QGF fixed .04 | 0.04 | - | - | - | 1 / 8 | - |
| 5 / 23 | QGF tuned | 40 | - | - | - | 1 / 8 | - |
| 5 / 23 | IPG | 5 | - | - | - | 8 / 8 | 0 / 0.01 |
| 5 / 23 | FK corrector | 5 | - | - | - | 16 / 32 | 0 / - |
| 5 / 23 | DF Regular, raw Q, λ0 | 5 | 0.1 / 128 | 0 | - | 1 / 8 | - |

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

# Selected settings (64-84 of 84)

| Task / seed | Requested condition | κ | η / l | λ | σ / K | N / M | noise / ridge |
|---|---|---|---|---|---|---|---|
| 5 / 23 | DF Regular, raw Q, λ1 | 5 | 0.1 / 128 | 1 | - | 1 / 8 | - |
| 5 / 23 | DF Stochastic, raw Q, λ0 | 5 | 0.1 / 128 | 0 | - | 1 / 8 | - |
| 5 / 23 | DF Stochastic, raw Q, λ1 | 40 | 0.01 / 2 | 1 | - | 1 / 8 | - |
| 5 / 23 | DF Regular, average Q, λ0 | 80 | 0.1 / 8 | 0 | 0.01 / 16 | 1 / 8 | - |
| 5 / 23 | DF Regular, average Q, λ1 | 5 | 0.1 / 128 | 1 | 0.01 / 16 | 1 / 8 | - |
| 5 / 23 | DF Stochastic, average Q, λ0 | 1 | 0.1 / 256 | 0 | 0.01 / 16 | 1 / 8 | - |
| 5 / 23 | DF Stochastic, average Q, λ1 | 40 | 0.01 / 2 | 1 | 0.01 / 16 | 1 / 8 | - |
| 5 / 37 | BC flow | 0 | - | - | - | 1 / 8 | - |
| 5 / 37 | QGF fixed .02 | 0.02 | - | - | - | 1 / 8 | - |
| 5 / 37 | QGF fixed .04 | 0.04 | - | - | - | 1 / 8 | - |
| 5 / 37 | QGF tuned | 40 | - | - | - | 1 / 8 | - |
| 5 / 37 | IPG | 8 | - | - | - | 8 / 32 | 0 / 0.01 |
| 5 / 37 | FK corrector | 40 | - | - | - | 16 / 8 | 0.02 / - |
| 5 / 37 | DF Regular, raw Q, λ0 | 5 | 0.1 / 128 | 0 | - | 1 / 8 | - |
| 5 / 37 | DF Regular, raw Q, λ1 | 80 | 0.1 / 16 | 1 | - | 1 / 8 | - |
| 5 / 37 | DF Stochastic, raw Q, λ0 | 80 | 0.1 / 8 | 0 | - | 1 / 8 | - |
| 5 / 37 | DF Stochastic, raw Q, λ1 | 80 | 0.1 / 8 | 1 | - | 1 / 8 | - |
| 5 / 37 | DF Regular, average Q, λ0 | 80 | 0.1 / 8 | 0 | 0.2 / 4 | 1 / 8 | - |
| 5 / 37 | DF Regular, average Q, λ1 | 80 | 0.03 / 32 | 1 | 0.2 / 4 | 1 / 8 | - |
| 5 / 37 | DF Stochastic, average Q, λ0 | 80 | 0.1 / 8 | 0 | 0.03 / 16 | 1 / 8 | - |
| 5 / 37 | DF Stochastic, average Q, λ1 | 80 | 0.1 / 8 | 1 | 0.03 / 4 | 1 / 8 | - |

N=particles; M=flow time slices. A dash means unused or still pending, not zero. Stochastic D-Flow uses T=1; particle noise is a separate setting. FK does not use IPG ridge. Complete configuration dictionaries and selection states are in SELECTED_SETTINGS.json. Individually selected new-Qavg settings may still await the all-three-model test barrier.

# Failures, limits and numerical checks

All six model math/profile gates completed. The preserved raw float32 finite-difference checks include 6 failures out of 774 raw rows. Separately declared reference criteria passed: 678 non-FD checks and 96 double-reference comparisons, with 0 qualified failures. No tolerance, checkpoint, production precision or numerical-source change was made to erase the raw failures.

Engineering profile: 84 settings, 168 offline sampler calls and 3360 action rows. These are not physical episodes. The largest observed steady B20 call was 0.371 seconds; this is not a full-episode runtime guarantee.

A source-description error was found and corrected before fresh40 release: actual reset IDs, not row indices 0-19, enter the source-key folding. Disjoint A/B reset IDs create 40 distinct episode-key contexts under a shared policy root. The old wording is preserved; numerical code and cohorts were unchanged.

A repeated canonical BC validation control can consume real compute while reproducing already observed trajectories. Such repeats count as executions/cost but not new independent trajectory evidence. Raw completed-record counters after a resume may include an earlier partial prefix; report unique completed evidence or subtract separately preserved prefix work before claiming incremental attempt cost.

Q averaging may suppress a narrow false peak or a genuine narrow optimum. Noise may help movement within the actor’s image or amplify a mistaken score. Native success alone does not identify which mechanism occurred. Learned-V Bellman residuals measure self-consistency, not ground-truth Q error. Saved traces do not contain every latent, perturbation bank or counterfactual Q value.

Source-only boundary note (0 new queries): the current positive-σ objective differentiates through clipping of perturbed actions, even though identity guidance bypasses the separate flow-endpoint Jacobian. For Q(a)=a at a=1 with offsets ±δ, 0<δ<2, the averaged score derivative is1/2 versus the raw derivative1. In multiple dimensions clip masks can also change direction; small positive σ need not recover a boundary gradient. This mathematical property is not a causal explanation of these outcomes. A prospective fixed raw/Qavg strength control remains unimplemented and unrun; no current method, setting or comparison changes.

Small validation panels and hierarchical search can miss useful settings or select noisy winners. Unequal inference iterations, particle/flow counts, early stopping and compilation confound runtime comparisons. One trained model cannot establish a universal method winner. All completed zeros remain in the result tables; pending and interrupted cells remain explicit.

The auxiliary panels below are separate diagnostic validation gates. They are not Task2 full-domain tuning, a new D-Flow comparison, or any part of the main Task1/5 target100 heldout sample.

# Auxiliary diagnostic: Task 2 competence

Status: COMPLETE_AUDITED. Diagnostic validation only; separate from all main heldout denominators. No training, new tuning or algorithm ranking.

Official medium-maze Task2 starts around (0,20) and aims for (20,0), with the original1000-control cap. The gate uses the exact frozen BC1k and Task2-specific IQL Q/V50k for models11/23/37. Four locked policies on8 fresh diagnostic starts per model request96 executions.

The tuned reference is locked historical validation-tuned QGF: κ12/40/20 for models11/23/37. It is not a new current-domain sweep including κ80, and does not establish future full-comparison QGF selection as complete. Older historical60/60 is readiness context, not a harder-task or future100 guarantee.

| Diagnostic condition | Seed11 | Seed23 | Seed37 |
|---|---|---|---|
| BC | 0/8 | 0/8 | 0/8 |
| QGF fixed .02 | 0/8 | 0/8 | 0/8 |
| QGF fixed .04 | 0/8 | 0/8 | 0/8 |
| QGF historical VAL-tuned | 8/8 | 8/8 | 7/8 |

Observed completed diagnostic validation executions: 96 of96 requested. Invalid action/native steps: 0; nonfinite-return episodes: 0. All zeros and failures are retained; unfinished physical prefix cost is unknown.

Saved-data audit:
outputs/overnight_research_20261106_night/review/task2_competence_completed/20261107T070757_505892Z/REVIEW.json
SHA256 231ffe107c2dd5924b4d7b42f30a4f470c4f0ca774e35fa3c257eaca5d0f2d9f

# Auxiliary diagnostic: local coverage topology

Status: COMPLETE_AUDITED. Diagnostic validation only; separate from all main heldout denominators. No training, new tuning or algorithm ranking.

This separate stress diagnostic changes the reset distribution to fixed logged-TRAIN positions around the coverage region and uses a128-control cap. Upstream/downstream strata are constructed and frozen before model queries; goal/topology follow the declared native maze. It is not the standard1000-control benchmark.

Three trained models each supply FULL/HOLE critic arms. The hole withholds a rectangular upward-action subset only from Q/V training; both arms keep the same full-data BC actor. BC and fixedκ40 QGF are evaluated on4 starts per stratum, two strata and two arms:96 actual requests. BC arm repeats and paired starts do not add independently trained models. No true-Q calibration or causal coverage gain is established.

| Diagnostic condition | Seed11 | Seed23 | Seed37 |
|---|---|---|---|
| upstream / FULL / BC | 0/4 | 0/4 | 0/4 |
| upstream / FULL / QGF fixed40 | 4/4 | 2/4 | 4/4 |
| upstream / HOLE / BC | 0/4 | 0/4 | 0/4 |
| upstream / HOLE / QGF fixed40 | 4/4 | 4/4 | 4/4 |
| downstream / FULL / BC | 0/4 | 0/4 | 0/4 |
| downstream / FULL / QGF fixed40 | 4/4 | 4/4 | 4/4 |
| downstream / HOLE / BC | 0/4 | 0/4 | 0/4 |
| downstream / HOLE / QGF fixed40 | 4/4 | 4/4 | 4/4 |

Observed completed diagnostic validation executions: 96 of96 requested. Invalid action/native steps: 0; nonfinite-return episodes: 0. All zeros and failures are retained; unfinished physical prefix cost is unknown.

Saved-data audit:
outputs/overnight_research_20261106_night/review/coverage_topology_gate_completed/20261107T072144_992144Z/REVIEW.json
SHA256 13030dd477c668a3369a50621c150ba4f481fe8b0d0f378a3903fd979b7a1410

# Expanded QGF validation: separate FULL / HOLE follow-up

Status: COMPLETE_AUDITED. This is a separate validation-only follow-up on newly reserved, logged-TRAIN reset positions with a128-control cap. It is not the standard Task1/5 heldout target100 panel and adds no heldout evidence to its denominators.

The independently declared26-strength grid is κ=0,.02,.04,.06,.08,.1,.2,.3,.4,.6,.8,1,2,3,5,8,10,12,16,20,32,40,64,80,160,320. Every strength receives the same balanced8 validation starts:4 upstream and4 downstream, independently for each of six critic-arm/model cells. Complete execution is156 B8 batches /1,248 actual validation trials. κ0 is the canonical BC behavior for this QGF kernel.

| Model / critic arm | Own κ | VAL /8 | Up /4 | Down /4 | Positive κ |
|---|---|---|---|---|---|
| 11 / FULL | 64 | 8/8 | 4/4 | 4/4 | 64 |
| 11 / HOLE | 40 | 8/8 | 4/4 | 4/4 | 40 |
| 23 / FULL | 32 | 6/8 | 2/4 | 4/4 | 32 |
| 23 / HOLE | 40 | 8/8 | 4/4 | 4/4 | 40 |
| 37 / FULL | 80 | 8/8 | 4/4 | 4/4 | 80 |
| 37 / HOLE | 40 | 8/8 | 4/4 | 4/4 | 40 |

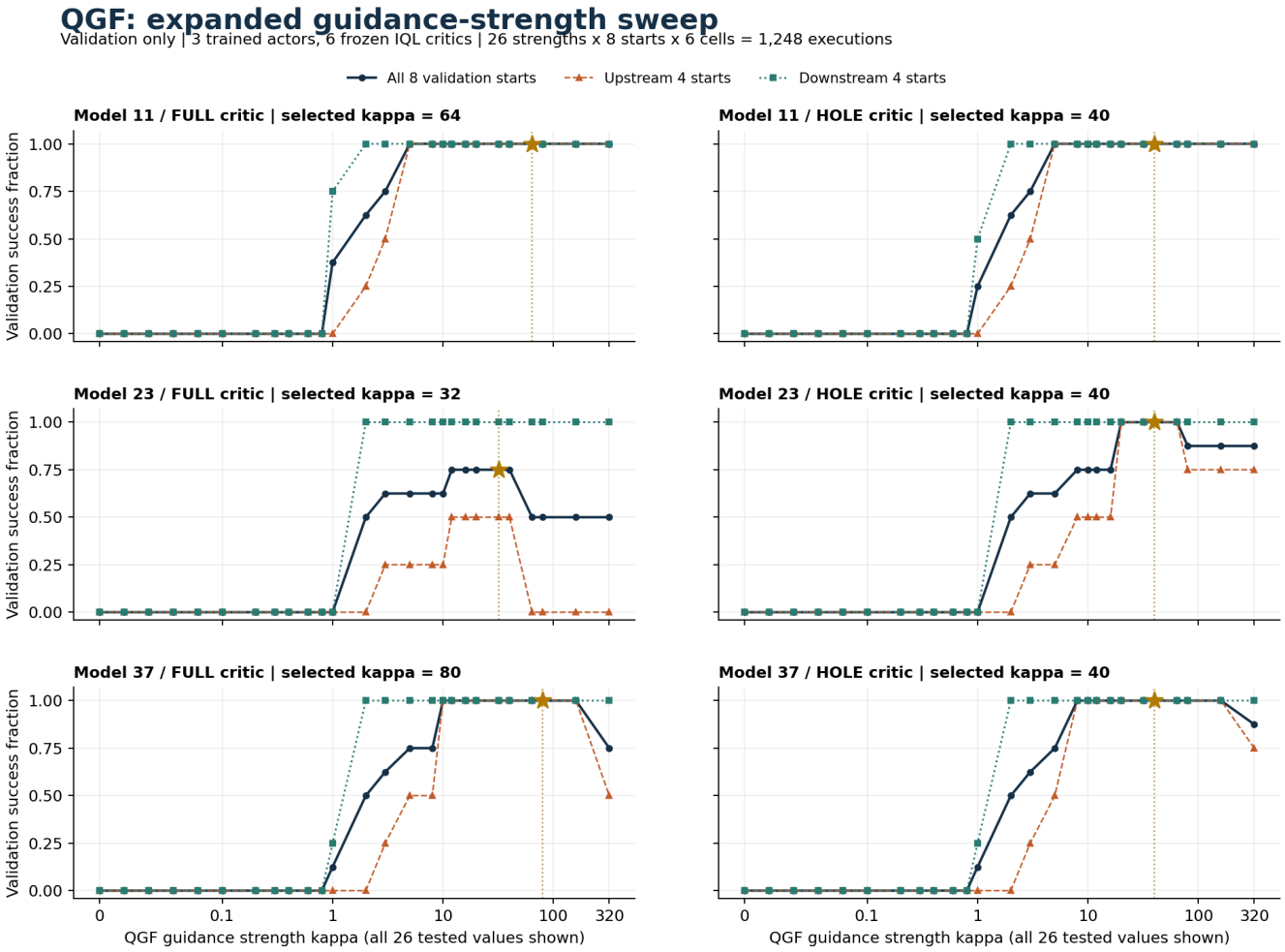
Observed completed validation: 156 B8 batches / 1248 actual episodes. Completed zero-success batches: 68; completed invalid action steps: 0; completed nonfinite-return episodes: 0. All candidate outcomes remain in the bound raw evidence, including zeros and failures; unfinished prefix outcomes and cost are unknown.

Saved-data audit:
outputs/overnight_research_20261106_night/review/tuned_topology_qgf26_completed/20261107T075453_232660Z/REVIEW.json
SHA256 22f29f3a3847a9cafb8b3451a4336ac4522a30fd875745b29c20910ecc2417d1

Three trained BC actors are shared across FULL/HOLE critic arms. The rectangular upward-action hole affects only Q/V training; both arms retain the full-data BC actor. Each arm chooses its own validation setting, so differences combine critic coverage and tuning. Validation success does not establish true-Q calibration, a heldout winner or a causal benefit from deleting data.

The proposed D-Flow extension retains all eight identity/full-flow conditions with independently fixed λ0/1, expanded I through512 and Qavg strata. The actual I512 engineering outcome is reported separately below. Expanded D-Flow/IPG/FK validation, the all66-guided-choice global freeze and future100 tests remain unexecuted. The six-QGF freeze is explicitly not that full-comparison freeze; this appendix authorizes no further queries.

# All 26 QGF strengths: saved validation curves




All26 predeclared strengths are displayed for each of six critic-arm/model cells. Curves show the same8 diagnostic validation starts, including4 upstream and4 downstream; stars identify each cell's already-frozen own validation choice. The figure is generated only from the audited saved scalars and adds0 trials, model queries or tuning decisions. It is not heldout evidence or part of the main100.

# Actual I512 engineering gate: separate outcome

Status: FAILED_AUDITED. This is an offline numerical gate, not native validation or a heldout method comparison. It adds0 training updates and0 physical episodes. The main I≤256 qualified gates and frozen comparison settings remain unchanged.

| Critic cell | Gate | Non-FD fail / checks | Raw FD32 fail / rows | Eligible FD64 pass / NA |
|---|---|---|---|---|
| seed11_full | FAIL | 4 / 117 | 4 / 48 | 44 / 4 |
| seed11_hole | FAIL | 4 / 117 | 3 / 48 | 46 / 2 |
| seed23_full | FAIL | 5 / 117 | 4 / 48 | 46 / 2 |
| seed23_hole | FAIL | 4 / 117 | 4 / 48 | 46 / 2 |
| seed37_full | FAIL | 5 / 117 | 3 / 48 | 46 / 2 |
| seed37_hole | FAIL | 1 / 117 | 4 / 48 | 46 / 2 |

Non-FD checks passed 679/702; 23 remained failed. Maximum failed-check absolute difference: 5.7220459e-06; declared failed-check tolerance(s): 2e-06. No threshold waiver or changed numerical source is used to relabel these failures.

Raw float32 finite differences retain 22 failures in 288 rows. Separately, 274 eligible double-reference FD rows passed, 14 nonsmooth rows were NA, and all 288 production-AD32/reference-AD64 comparisons passed. Those checks do not cancel the failed non-FD update checks.

Observed profiles: 84 rows; largest repeated same-shape B20 second-call time 0.950311s. B8/B16 fields are single observed calls and can include first-shape compilation for raw delegates; they are not guaranteed steady timings or full-rollout bounds.

Expanded DF prescreen, validation and tests remain blocked by the failed engineering gate. The saved-artifact audit passed verification of the failed result; it did not qualify the implementation. The original failing gate did not save its one-step operand vectors. Later diagnostic captures are reported separately below; compilation/fusion/rounding is only a possible explanation and a gradient issue cannot be excluded. Any new qualification requires a separate declared release.

All six full-agent trees and bound source/weight bytes are unchanged. Saved-data audit:
outputs/overnight_research_20261106_night/review/I512_engineering_completed/20261107T083628_677851Z/REVIEW.json
SHA256 f9bdafc8e98e6088130db2b0040b76b50b4ea5445527404869fa3b6d8d06bcf9

# One-step operand diagnostic: descriptive follow-up

Status: COMPLETE_AUDITED. This separately declared follow-up captures numerical operands; it does not qualify, modify or waive the original I512 engineering FAIL. It adds 0 native episodes, training updates or main heldout samples.

| Critic cell | Configuration cases | Literal fail κ320 / κ80 | Conditional interval compatible / outside / NA |
|---|---|---|---|
| seed11_full | 16 | 4 / 0 | 16 / 0 / 0 |
| seed11_hole | 16 | 4 / 0 | 16 / 0 / 0 |
| seed23_full | 16 | 5 / 0 | 16 / 0 / 0 |
| seed23_hole | 16 | 4 / 0 | 16 / 0 / 0 |
| seed37_full | 16 | 5 / 0 | 16 / 0 / 0 |
| seed37_hole | 16 | 1 / 0 | 16 / 0 / 0 |

The fixed panel contains 96 sampler-configuration cases over repeated four-anchor TRAIN engineering panels: regular/stochastic Qavg identity guidance, K4/16, λ0/1 and κ80/320, with η=.1, σ=.2 and one source update. It is not a new I512 rollout sweep or 96 independent environment trials.

The unchanged literal 2e-6 update test still fails in 23 cases: 23 at κ320 and 0 at κ80. Failed capture invariants: 0; public source/action AD32-versus-AD64 disagreement cases beyond the declared AD tolerance: 0/0. All raw failures and case flags remain saved.

The conditional inferred-gradient interval, compared using 2e-5 + 2e-4|AD64|, was compatible in 96 cases, outside in 0, and NA in 0. This includes a gradient-disagreement allowance and does not isolate update rounding. The arithmetic model is not proved for the compiled production graph. Public gradients use a separate compiler context; the internal production-loop gradient was not directly captured. Compatibility does not establish the cause or exclude a gradient issue.

All six frozen full-agent states and bound source/weight bytes are unchanged. COMPLETE denotes acquisition of the fixed diagnostic panel only. The original engineering gate remains FAIL and the expanded prescreen/validation/test barrier remains closed.

Saved-data audit:
outputs/overnight_research_20261106_night/review/I512_one_step_discriminator_completed/20261107T085242_740806Z/REVIEW.json
SHA256 3eb5a91b5c281c8f3e8722da952d4891a7f60a3436adddbb596dfa3a87bc1c97

# Known-linear arithmetic control: descriptive addendum

A separate zero-policy, known-linear-critic fixture completed 32 configurations over four repeated source contexts (128 represented rows, 0 learned-model queries, training updates or native episodes). All 32 raw literal 2e-6 update comparisons passed at κ80/320. This negative control did not reproduce or explain the learned-model 23 failures and does not qualify the original engineering gate.

Exact public-gradient-versus-stored-β invariants failed in 8 configurations; these remain failures and their conditional arithmetic bounds remain NA. Conditional bounds were within in 24, outside in 0, and NA in 8. The maximum public-gradient coefficient difference was 2.98e-8. Production-loop gradients were not directly observed; the externally prescribed stochastic sample matching the compiled production noise remains an unproved assumption. The saved JSON/hash audit did not independently replay NPZ arrays or establish the cause of the learned-model failures. The original I512 gate remains FAIL.

Synthetic-control saved-data audit:
outputs/overnight_research_20261106_night/review/known_linear_control_completed/20261107T092002Z/REVIEW.json
SHA256 35790b552529645802413f6e125f9798efb175acff392fca593068d4df992acd

# Task 2: offline engineering and preserved failures

This separate TRAIN-only engineering attempt acquired all three frozen Task2 BC1k/IQL50k agents. It made 0 native environment steps, validation/test episodes or training updates. It does not add to the main Task1/5 target100 sample, select full-domain Task2 settings or waive the failed coverage I512 gate.

| Model | Qualified math / profile | Raw FD32 failures /16 | FD64 pass / NA | B20 second-call max (s) |
|---|---|---|---|---|
| 11 | PASS / PASS | 1 /16 | 16 / 0 | 0.146694 |
| 23 | PASS / PASS | 0 /16 | 16 / 0 | 0.149431 |
| 37 | PASS / PASS | 2 /16 | 16 / 0 | 0.146116 |

All 339 non-FD checks and 48 production-AD32/reference-AD64 checks passed. All 48 eligible FD64 comparisons passed with 0 NA; the 3 raw float32 finite-difference failures remain saved (models11/23/37:1/0/2). The attempt recorded 51 setting profiles, 102 offline sampler calls and 2040 action rows. Two fixed TRAIN observations were repeated ten times for B20 with distinct source IDs; these are not 20 independent states or native episodes.

Each setting was called twice; first calls can include compilation, and second-call times are single measured B20 batch costs, counted once rather than multiplied by 20. These probes do not measure native episode lengths, exhaustive-grid cost or a full-study budget. All full-agent component trees and bound source/weight bytes were unchanged.

Preserved engineering failures: the V1 worker stopped before restoring any model because its byte-only TRAIN hash was compared with a dtype+shape-prefixed reference hash. An independent root data check confirmed unchanged data under the correct convention. A separate V2 supervisor authorization was rejected before any child/model query because its required review map was empty. Both records remain intact; the later explicitly released V2 attempt repaired the hash convention and supplied the review bindings, without changing observations, models, samplers, settings or tolerances.

The earlier 96-episode Task2 competence gate remains diagnostic validation. A separate completed locked-control fresh100 panel follows; it preserves historical QGF strengths 12/40/20. A full current-domain Task2 D-Flow/QGF/IPG/FK study and its own all-three-model selection freeze remain pending and unpriced.

Saved-data audits:
outputs/overnight_research_20261106_night/review/task2_math_profile_v2_completed/20261107T105853Z/REVIEW.json
SHA256 b9df35b8c923a3532abb4faa2afecfac2ee9ce5a121169db91ed17b0b0869979
Independent:
outputs/overnight_research_20261106_night/review/task2_v2_independent_completed/20261107T110733_447945Z/REVIEW.json
SHA256 c847404a5cfc566b87b6c8be613fa1b9e27fc2a786cfd60ef3b898e124a9cdfc

# Task2 locked controls: fresh 100-episode heldout panel

All three frozen BC1k / Task2-specific IQL50k models were evaluated on the same 100 newly reserved task-native reset contexts. Task2 starts near (0,20) and aims for (20,0), using actual next states and the unchanged 1000-control cap. All twelve settings froze before any of these tests. No new training, strength selection or D-Flow sweep was performed.

| Locked policy | Model11 /100 | Model23 /100 | Model37 /100 |
|---|---|---|---|
| BC flow | 0/100 | 0/100 | 0/100 |
| QGF fixed .02 | 0/100 | 0/100 | 0/100 |
| QGF fixed .04 | 0/100 | 0/100 | 0/100 |
| QGF locked historical validation-tuned | 100/100 | 100/100 | 92/100 |

The locked historical validation-tuned strengths are κ12, κ40 and κ20 for models11,23,37. These goal-competence results retain all nine zero control cells. They do not complete the latest 20-strength tuning domain or establish a D-Flow comparison. Same reset contexts across models/methods are paired; there are three independently trained models, not 1200 independent models.

This separate study executed 60 physical B20 batches / 1200 new heldout episodes, 954127 physics steps and 954127 policy calls. Sampler time 98.893s and batch wall time 165.207s are counted once per batch. All model states and source bytes remained unchanged; invalid/nonfinite outcomes, aliases and unfinished prefixes were zero. The earlier 96 diagnostic episodes and the main Task1/5 8400 remain separate.

Saved-only audits:
outputs/overnight_research_20261106_night/review/task2_locked_controls100_completed/20261107T1203Z/REVIEW.json
SHA256 2006c6b285de028557fe8427267806c11d481b6bd7fe5800a30ecbb3f986fdaa
Independent: outputs/overnight_research_20261106_night/review/task2_locked100_independent_completed/REVIEW.json
SHA256 ebd919e2e32f750410713c7a01aac1b70d5b029fbbb91129c1eac71703fce586

# Main100 saved traces: posthoc action and source diagnostics

This analysis describes all 8400 already executed main episode rows and 420 physical B20 batches, covering all 84 method/model/task cells. It adds 0 trials, learned-Q calls, training updates or retuning. All 8400 first-frame pairing flags passed the saved context checks. The output audit verified scalar arithmetic and byte provenance; it did not independently replay the trajectory arrays.

# Fixed descriptive example: Task5, trained model23

| Policy | Success /100 | Action saturation | First action change vsBC | First raw-Q change vsBC |
|---|---|---|---|---|
| QGF fixed .02 | 0/100 | 23.1% [100/0] | 0.00145 [100/0] | 0.000134 [100/0] |
| QGF fixed .04 | 0/100 | 23.2% [100/0] | 0.00291 [100/0] | 0.000267 [100/0] |
| QGF tuned | 24/100 | 100.0% [100/0] | 1.35 [100/0] | 0.116 [100/0] |
| DF Regular, raw Q, λ0 | 26/100 | 100.0% [100/0] | 1.4 [100/0] | 0.119 [100/0] |
| DF Regular, raw Q, λ1 | 62/100 | 69.6% [100/0] | 0.779 [100/0] | 0.0261 [100/0] |
| DF Regular, average Q, λ1 | 62/100 | 31.8% [100/0] | 0.779 [100/0] | 0.0261 [100/0] |

Brackets show [finite/NA] episode counts. Saturation is the per-episode fraction of executed actions with any coordinate |a|>.999, then averaged across episodes. The tuned QGF and raw regular λ0 rows saturate throughout these trajectories; raw regular λ1 and averaged regular λ1 differ despite equal success counts. These two λ1 rows share κ=5, η=.1 and I=128; their raw/averaged score implementation differs. Other rows use their own selected settings. Saturation differences do not establish critic repair or explain success. These posthoc associations also mix visited states and episode duration.

The .02/.04 QGF controls make only tiny first-action changes relative to paired BC in this example. Their zero successes do not imply a missing per-slice refresh: faithful original QGF recomputes its correction at all eight slices. Stochastic source search and Q averaging operate on the frozen actor/critic; they add neither offline transition data nor exploratory critic learning.

A larger first raw-Q score means the same frozen target-min IQL critic preferred the selected action at a paired initial state. It is not true-Q calibration, improvement of Q during source optimization, or proof of better unseen-action values. IPG/FK chosen-action Q was not saved and remains NA. For raw BC/QGF/DF, final_qavg_q is a copy of raw Q rather than an actual neighbor average.

# Saved source movements: each trained model

Only averaged DF saved the source fields below; raw DF/BC/QGF source fields are unavailable. Each entry is the mean over 100 episodes of a saved first-action-generation scalar. Brackets show [finite/NA] episode counts. Full source, score, saturation and particle summaries for every method and trained model remain in the bound all-model CSV.

| Task / model | Condition | Net ‖zl-z0‖ | Mean ‖κg‖ | Mean ‖λz‖ | Mean noise increment |
|---|---|---|---|---|---|
| 1 / 11 | DF Regular, average Q, λ0 | 32.8 [100/0] | 1.37 [100/0] | 0 [100/0] | 0 [100/0] |
| 1 / 11 | DF Regular, average Q, λ1 | 5.13 [100/0] | 6.75 [100/0] | 4.84 [100/0] | 0 [100/0] |
| 1 / 11 | DF Stochastic, average Q, λ0 | 19.5 [100/0] | 5.99 [100/0] | 0 [100/0] | 0.303 [100/0] |
| 1 / 11 | DF Stochastic, average Q, λ1 | 5.59 [100/0] | 6.16 [100/0] | 5.08 [100/0] | 0.304 [100/0] |
| 1 / 23 | DF Regular, average Q, λ0 | 25.9 [100/0] | 1.11 [100/0] | 0 [100/0] | 0 [100/0] |
| 1 / 23 | DF Regular, average Q, λ1 | 2.33 [100/0] | 2.65 [100/0] | 1.97 [100/0] | 0 [100/0] |
| 1 / 23 | DF Stochastic, average Q, λ0 | 55.1 [100/0] | 4.01 [100/0] | 0 [100/0] | 0.555 [100/0] |
| 1 / 23 | DF Stochastic, average Q, λ1 | 2.77 [100/0] | 3.7 [100/0] | 2.08 [100/0] | 0.303 [100/0] |
| 1 / 37 | DF Regular, average Q, λ0 | 31.6 [100/0] | 2.53 [100/0] | 0 [100/0] | 0 [100/0] |
| 1 / 37 | DF Regular, average Q, λ1 | 4.81 [100/0] | 5.03 [100/0] | 4.34 [100/0] | 0 [100/0] |
| 1 / 37 | DF Stochastic, average Q, λ0 | 19.6 [100/0] | 5.21 [100/0] | 0 [100/0] | 0.303 [100/0] |
| 1 / 37 | DF Stochastic, average Q, λ1 | 5.32 [100/0] | 5.43 [100/0] | 4 [100/0] | 0.303 [100/0] |
| 5 / 11 | DF Regular, average Q, λ0 | 2.02 [100/0] | 0.0795 [100/0] | 0 [100/0] | 0 [100/0] |
| 5 / 11 | DF Regular, average Q, λ1 | 1.4 [100/0] | 0.489 [100/0] | 0.503 [100/0] | 0 [100/0] |
| 5 / 11 | DF Stochastic, average Q, λ0 | 4.24 [100/0] | 2.95 [100/0] | 0 [100/0] | 0.125 [100/0] |
| 5 / 11 | DF Stochastic, average Q, λ1 | 4.2 [100/0] | 5.59 [100/0] | 2.85 [100/0] | 0.125 [100/0] |
| 5 / 23 | DF Regular, average Q, λ0 | 4.26 [100/0] | 5.37 [100/0] | 0 [100/0] | 0 [100/0] |
| 5 / 23 | DF Regular, average Q, λ1 | 1.36 [100/0] | 0.467 [100/0] | 0.494 [100/0] | 0 [100/0] |
| 5 / 23 | DF Stochastic, average Q, λ0 | 9.21 [100/0] | 0.0584 [100/0] | 0 [100/0] | 0.559 [100/0] |
| 5 / 23 | DF Stochastic, average Q, λ1 | 0.253 [100/0] | 3.39 [100/0] | 1.32 [100/0] | 0.176 [100/0] |
| 5 / 37 | DF Regular, average Q, λ0 | 4.09 [100/0] | 5.15 [100/0] | 0 [100/0] | 0 [100/0] |
| 5 / 37 | DF Regular, average Q, λ1 | 3.12 [100/0] | 5 [100/0] | 2.1 [100/0] | 0 [100/0] |
| 5 / 37 | DF Stochastic, average Q, λ0 | 4.43 [100/0] | 5.33 [100/0] | 0 [100/0] | 0.555 [100/0] |
| 5 / 37 | DF Stochastic, average Q, λ1 | 3.19 [100/0] | 5.5 [100/0] | 2.24 [100/0] | 0.555 [100/0] |

κg and λz norms omit η. The stochastic noise-increment norm includes sqrt(2ηT), T=1; regular has no added outer ξ. Means of norms do not reveal direction, cancellation or a per-iteration Q trajectory. Later active-call means in the CSV mix policy visitation and termination; they are not matched-state causal comparisons. Particle ESS/correction norms are ensemble/flow-slice diagnostics and cannot be substituted

for DF source drift or chosen-action Q.

All 84 ×38 metric summaries, including finite/NA denominators: outputs/overnight_research_20261106_night/main100_trace_diagnostic/runs/main100-saved-trace-de2e2e5258cd47a8834fbdf6c57710b3/PER_MODEL_METRIC_SUMMARY.csv
SHA256 67c0e41e23dbcd708b0b2f4c4c2cd729b9e03656d0535cd55d12748988bda426
Saved episode rows: outputs/overnight_research_20261106_night/main100_trace_diagnostic/runs/main100-saved-trace-de2e2e5258cd47a8834fbdf6c57710b3/EPISODE_METRICS.csv
SHA256 22cfbc94ed96d05623d9278274554b7f6904b5b7a2b6f1978fc9b1caa4e2a844

Saved-scalar audit:
outputs/overnight_research_20261106_night/review/main100_trace_completed/20261107T112322_652338Z/REVIEW.json
SHA256 35f10339aa39daeb33e5e8440c1735b26ce513b97f3ae02c68690e7b3c57bdeb

# Fixed-continuation critic reference: raw first actions

Status: COMPLETE_AUDITED. This separately declared privileged simulator diagnostic uses two canonical TRAIN states nearest nominal (12,4) and (12,8). It is not historical state replay or heldout navigation generalization. It does not tune any sampler or critic and does not add to the 10200 heldout episodes.

The first action is forced to (0,+1) or (0,-1). Afterwards, one fixed full-data QGF continuation policy πref is used for both FULL and HOLE critic scores: κ64/32/80 for models11/23/37, with the faithful original eight slices. The observed return belongs to this continuation, not an optimal policy or the IQL implicit policy. Predictions-minus-returns are πref-relative discrepancies, neither Q* errors nor exact IQL estimation errors.

| Model / anchor / first action | View | Q FULL / HOLE | MC mean ± descriptive SE | Q − MC FULL / HOLE |
|---|---|---|---|---|
| 11 / 0 / up | raw | -82.04 / -83.64 | -57.42 ± 0.038 | -24.62 / -26.22 |
| 11 / 0 / down | raw | -82.90 / -84.40 | -58.29 ± 0.000 | -24.61 / -26.11 |
| 11 / 1 / up | raw | -73.32 / -73.37 | -47.97 ± 0.038 | -25.35 / -25.40 |
| 11 / 1 / down | raw | -73.90 / -73.90 | -49.00 ± 0.000 | -24.90 / -24.90 |
| 23 / 0 / up | raw | -83.03 / -84.02 | -59.10 ± 0.021 | -23.93 / -24.92 |
| 23 / 0 / down | raw | -83.72 / -84.68 | -59.91 ± 0.020 | -23.81 / -24.77 |
| 23 / 1 / up | raw | -73.48 / -73.65 | -49.99 ± 0.025 | -23.49 / -23.66 |
| 23 / 1 / down | raw | -74.23 / -74.63 | -50.94 ± 0.041 | -23.29 / -23.69 |
| 37 / 0 / up | raw | -82.33 / -83.28 | -58.56 ± 0.265 | -23.77 / -24.72 |
| 37 / 0 / down | raw | -83.11 / -84.04 | -58.99 ± 0.181 | -24.12 / -25.06 |
| 37 / 1 / up | raw | -72.74 / -72.84 | -49.04 ± 0.232 | -23.69 / -23.80 |
| 37 / 1 / down | raw | -73.61 / -73.83 | -50.24 ± 0.341 | -23.37 / -23.59 |

Both critics rank the forced up action above down, matching observed paired Monte Carlo sample-mean ordering in all 12 raw critic-arm comparisons over six underlying model/anchor pairs. Including the two overlapping σ views gives 36/36 observed agreements; shared returns and keys do not create independent comparisons or guarantee true expected-return rankings.

An additive constant in Q would leave its action gradients and rankings unchanged. Negative πref-relative offsets therefore do not themselves demonstrate damaged guidance; these sparse forced-action observations neither measure the full action gradient nor establish an unseen-Q wrong-rank failure.

Anchor0 is near nominal (12,4); anchor1 is near (12,8). All twenty predeclared common keys remain in each return mean. The same trajectory returns are scored by both critic arms, so the two scores do not create additional episodes. Every model, anchor, action and raw/neighbor view remains in the bound all-model scalar CSV.

The diagnostic constructed new canonical native states from logged TRAIN tuples; raw qpos/qvel, goal, integration state, wrapper state, banks and continuation keys froze before any model/Q query. The initial engineering failure at goal-container copying remains preserved; the separately reviewed V2 copies that goal explicitly without changing the frozen numerical sampler. Prefix checks and internal reset steps are separate construction/engineering work.

# Matched nearby-action averages and fixed-policy returns

Each σ view uses the same four antithetic standard-normal offsets for the finite-bank score and its Monte Carlo target. The score is mean_k min_h Q_h(s,clip(a+σεk)); the matched target is the same weighted mean of forced-neighbor returns under πref. Exact clipped-action aliases preserve their 1/4 weights. Raw scores are compared with raw-action returns; averaged scores are compared with these matched neighbor targets, not the raw nominal target.

| Model / anchor / first action | View | Qbar FULL / HOLE | Matched MC mean ± SE | Qbar − MC FULL / HOLE |
|---|---|---|---|---|
| 11 / 0 / up | σ.03 | -82.05 / -83.64 | -57.44 ± 0.032 | -24.61 / -26.21 |
| 11 / 0 / up | σ.2 | -82.08 / -83.67 | -57.44 ± 0.028 | -24.63 / -26.23 |
| 11 / 0 / down | σ.03 | -82.89 / -84.39 | -58.28 ± 0.005 | -24.61 / -26.11 |
| 11 / 0 / down | σ.2 | -82.87 / -84.37 | -58.23 ± 0.014 | -24.64 / -26.14 |
| 11 / 1 / up | σ.03 | -73.32 / -73.37 | -47.92 ± 0.038 | -25.40 / -25.45 |
| 11 / 1 / up | σ.2 | -73.33 / -73.38 | -47.95 ± 0.033 | -25.38 / -25.43 |
| 11 / 1 / down | σ.03 | -73.90 / -73.90 | -48.97 ± 0.016 | -24.93 / -24.93 |
| 11 / 1 / down | σ.2 | -73.89 / -73.89 | -48.94 ± 0.024 | -24.94 / -24.94 |
| 23 / 0 / up | σ.03 | -83.03 / -84.02 | -59.11 ± 0.010 | -23.93 / -24.91 |
| 23 / 0 / up | σ.2 | -83.05 / -84.04 | -59.11 ± 0.010 | -23.95 / -24.93 |
| 23 / 0 / down | σ.03 | -83.71 / -84.68 | -59.82 ± 0.015 | -23.90 / -24.86 |
| 23 / 0 / down | σ.2 | -83.69 / -84.66 | -59.80 ± 0.017 | -23.89 / -24.86 |
| 23 / 1 / up | σ.03 | -73.48 / -73.65 | -50.00 ± 0.019 | -23.48 / -23.65 |
| 23 / 1 / up | σ.2 | -73.49 / -73.67 | -50.00 ± 0.013 | -23.49 / -23.67 |
| 23 / 1 / down | σ.03 | -74.23 / -74.62 | -50.89 ± 0.040 | -23.33 / -23.73 |
| 23 / 1 / down | σ.2 | -74.21 / -74.60 | -50.78 ± 0.029 | -23.42 / -23.82 |
| 37 / 0 / up | σ.03 | -82.34 / -83.28 | -58.54 ± 0.161 | -23.80 / -24.75 |
| 37 / 0 / up | σ.2 | -82.36 / -83.31 | -58.41 ± 0.157 | -23.95 / -24.90 |
| 37 / 0 / down | σ.03 | -83.10 / -84.04 | -59.13 ± 0.126 | -23.97 / -24.91 |
| 37 / 0 / down | σ.2 | -83.07 / -84.01 | -59.10 ± 0.123 | -23.97 / -24.91 |
| 37 / 1 / up | σ.03 | -72.74 / -72.84 | -49.13 ± 0.136 | -23.61 / -23.72 |
| 37 / 1 / up | σ.2 | -72.76 / -72.87 | -49.14 ± 0.138 | -23.62 / -23.73 |
| 37 / 1 / down | σ.03 | -73.61 / -73.83 | -50.08 ± 0.186 | -23.52 / -23.74 |
| 37 / 1 / down | σ.2 | -73.58 / -73.80 | -50.04 ± 0.198 | -23.54 / -23.76 |

These are finite-bank, fixed-continuation observations. A smaller absolute discrepancy here would not demonstrate global critic repair, a new stochastic policy, improved unseen-action Q or heldout success. Shared common random keys couple models, actions and neighbors; 2160 executed episodes are not claimed to be 2160 independent observations. The displayed SE is descriptive within each twenty-key target, not a simultaneous confidence interval across the table. The separately saved finite-horizon Hoeffding intervals are broad and individual only.

The nominal discount is .99 with the production float32 stored value; true absorbing goals have zero tail. Nonabsorbing 1000-step truncation retains its separately logged geometric tail bound. This acquisition does not waive the failed coverage I512 gate or change main I≤256 settings.

| Model | Actual MC episodes / B20 | Physics / policy calls | Sampler / batch wall (s) |
|---|---|---|---|
| 11 | 720 / 36 | 55403 / 54683 | 5.299 / 10.244 |
| 23 | 720 / 36 | 58200 / 57480 | 5.475 / 10.595 |
| 37 | 720 / 36 | 57042 / 56322 | 5.752 / 10.901 |

All three acquisitions completed 108 physical B20 / 2160 valid MC episodes; critic-arm score views and clipped aliases do not multiply physical cost. The forced first action is one physics transition without a learned-policy call. Tables count timing once per physical batch. Construction and 4-step replay engineering probes are retained separately in the manifest.

All 72 matched target rows: outputs/overnight_research_20261106_night/fixed_continuation_calibration_v2/analysis/all_three_20261107T1218Z/ALL_MATCHED_TARGETS.csv
SHA256 03e593f63f81f12a0de6c6611e9677572d2b705c8d89662814b1e1b85f0cde8a
Independent all-three audit:
outputs/overnight_research_20261106_night/review/calibration_all3_independent/20261107T121925_633585Z/REVIEW.json
SHA256 40b4f23025dfe88e7381019d396d4013b455f733766f7ff7c731fe86fc2fdc7b

# Reproducibility and preserved campaigns

The evidence JSON binds complete CSV/metadata/trace triples with SHA256, exact reset IDs, policy roots, model inputs and selected configurations. New physical evidence and inherited references are counted separately. Selection uses validation only; heldout outcomes never retune or choose a family configuration.

Native model construction combines the frozen BC policy with the frozen goal-specific IQL critic. The two-head target minimum is used for guidance. Shared model hashes and full-agent tree hashes are checked by released workers before/after execution. Local CPU only: at most two scientific workers, bounded 7200-second attempts, explicit reviewed checkpoint resumes, no automatic failure retries.

| Task / seed | BC checkpoint SHA prefix | IQL checkpoint SHA prefix |
|---|---|---|
| 1 / 11 | 45496ed77dd755a7 | 9bb2d24f10135780 |
| 1 / 23 | ce24a7f3c86bbfaa | e5e09273f762357a |
| 1 / 37 | 1a01ebdb0ed1d8a9 | 995c4299ccabef98 |
| 5 / 11 | 45496ed77dd755a7 | fc61e43f2794c8c4 |
| 5 / 23 | ce24a7f3c86bbfaa | d07c86d71ed81012 |
| 5 / 37 | 1a01ebdb0ed1d8a9 | d6ec49560a722a9f |

Verified common configuration: H=1, flow slices=8, two Q heads with minimum aggregation; discount=0.99, expectile=0.8, target τ=0.005, batch=256, learning rates=0.0003. Actor hidden layers=[64, 64]; Q/V hidden layers=[128, 128]. Full checkpoint paths, configuration dictionaries and SHA256 values are included in the bound manifest/reference bundle.

# Core numerical source hashes

work/overnight_research_20261105/samplers.py
a615c61d284cb256e1291a5ae71cc719b03614a9218f84a49e86dfa6a80105e9

work/overnight_research_20261106_night/identity_math/kernel.py
0f8d7a4519ba95cc3e8aafcfc3996e381408c9dbc44ac555cb0ccae3cab79d56

work/overnight_research_20261106_continuation/exploratory_identity_qavg_v2/kernel.py
7d7d10799f5f3b81aad15668568c477aaf95bffe35949c47b701ec753ba01ddc

Historical reports and archives remain separate. The earlier all-experiments report includes synthetic one-point diagnostics, custom sequential learned-BC/IQL navigation, modified FULL/HOLE coverage studies and MuJoCo cube manipulation. FULL/HOLE labels concern older offline-data coverage arms; they do not mean full-flow versus Euler endpoints here.

Final archive verification: pending the final morning integrity receipt; this preview is not final delivery.

The fixed-continuation appendix records one separately declared privileged diagnostic. A broader prospective study should freeze complete TRAIN/validation anchor states, supported and critic-hole first actions, one independently competent continuation policy $\pi_{ref}$, and fresh common rollout keys before simulator Monte Carlo evaluation. Compare predicted Q with the discounted return under that same continuation and report support membership, ranking differences and uncertainty. This measures discrepancy relative to $\pi_{ref}$; it is neither $Q^*$ nor exact IQL estimation error. Match the reward, discount, action horizon and terminal masks. At a nonabsorbing truncation, $|r| \le 1$ gives $|tail| \le \gamma^H/(1-\gamma)$: for nominal $\gamma=.99$, about27.6252 after128 steps and .00431712 after1000. A true absorbing goal has zero tail; a learned-V bootstrap adds value uncertainty. Preserve separate TRAIN/validation/test data and a fresh reviewed protocol; do not retune current heldout settings. Task2's full unchanged grid remains pending and unpriced; cheap offline call profiles cannot price native episode lengths or establish a full-study budget. This broader methodological recommendation adds no experiment or claim of exact rollout-Q error.

Bound primary-paper/source interpretation and horizon supplement: review/q_reference_semantics/Q_REFERENCE_NOTE.txt; PROVENANCE.json; INDEPENDENT_MATH_REVIEW.json; HORIZON_SUPPLEMENT.txt/json. Exact source hashes are in the build manifest.